16 (a) Show that the equation $x^3 - 3x^2 + 3 = 0$ has a solution between $x = 2$ and $x = 3$

$f(x) = x^3 - 3x^2 + 3.$

$f(2) = 8 - 12 + 3 = -1.$

1 mark
Evaluate endpoints

$f(3) = 27 - 27 + 3 = 3.$

Continuous polynomial changes sign, so a root lies in(2,3).

1 mark
Sign-change argument

(2)

(b) Show that the equation $x^3 - 3x^2 + 3 = 0$ can be rearranged to give $x = \sqrt[3]{3x^2 - 3}$

$x^3 - 3x^2 + 3 = 0 \rightarrow x^3 = 3x^2 - 3.$

Cube-root both sides: $x = \sqrt[3]{3x^2 - 3}$

1 mark
Correct rearrangement

(1)

(c) Starting with $x_0 = 2$, use the iteration formula $x_{n+1} = \sqrt[3]{3x_n^2 - 3}$ to find the value of x_2
Give your answer correct to 3 decimal places.

1 mark
Substitute initial2

First: cube root($3\times2^2 - 3$) = 2.080083823...

1 mark
First iteration2.080...

Second: cube root($3\times2.080083823...^2 - 3$)

= 2.153015141... → 2.153 (3 decimal places).

1 mark
Second iteration2.153

(3)

(Total for Question 16 is 6 marks)